

UNIT-5

Measurement of Force

The Mechanical Quantity which changes or tends to change the motion or shape of a body to which it is applied is called force. Force is a basic Engineering Parameter, the measurement of which can be done in many ways-

① Direct Methods -

It involves a direct comparison with a known gravitational force on a standard mass, say by a balance.

② Indirect Methods -

It involves the measurement of effect of force on a body, such as acceleration of a body of known mass subjected to a force.

Device To Measure force -

• Scale and Balances -

- Equal arm Balance,
- Unequal arm Balance
- Pendulum Scale

• Elastic force meter (Proving Ring)

• Load cells -

a) Strain Gauge load cell

b) Hydraulic load cell

c) Pneumatic load cell

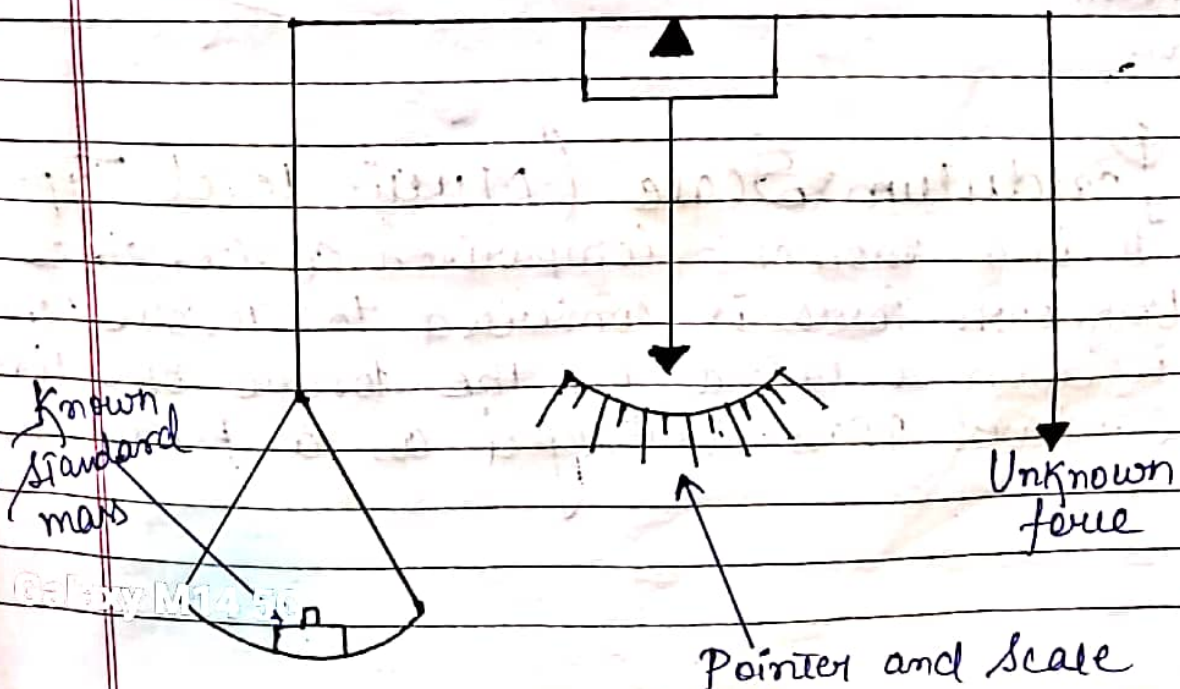
- Equal arm Balance - An Equal arm balance works on the Principle of moment comparison. The beam of the equal arm balance is in equilibrium position when,

Clockwise Rotating moment = Anti-Clockwise Rotating moment.

$$M_2 L_2 = M_1 L_1$$

i.e., the Unknown force is balanced against the known gravitational force.

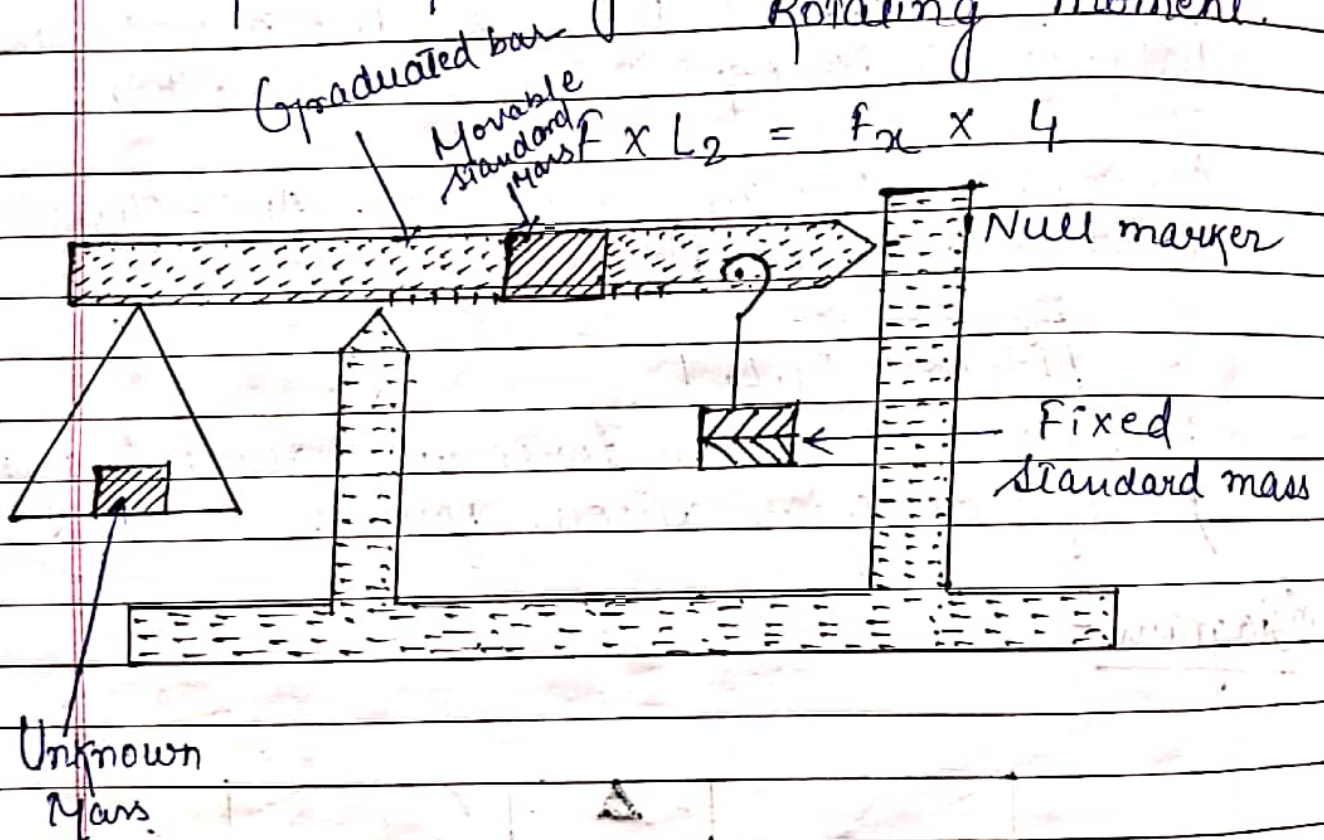
Diagram -



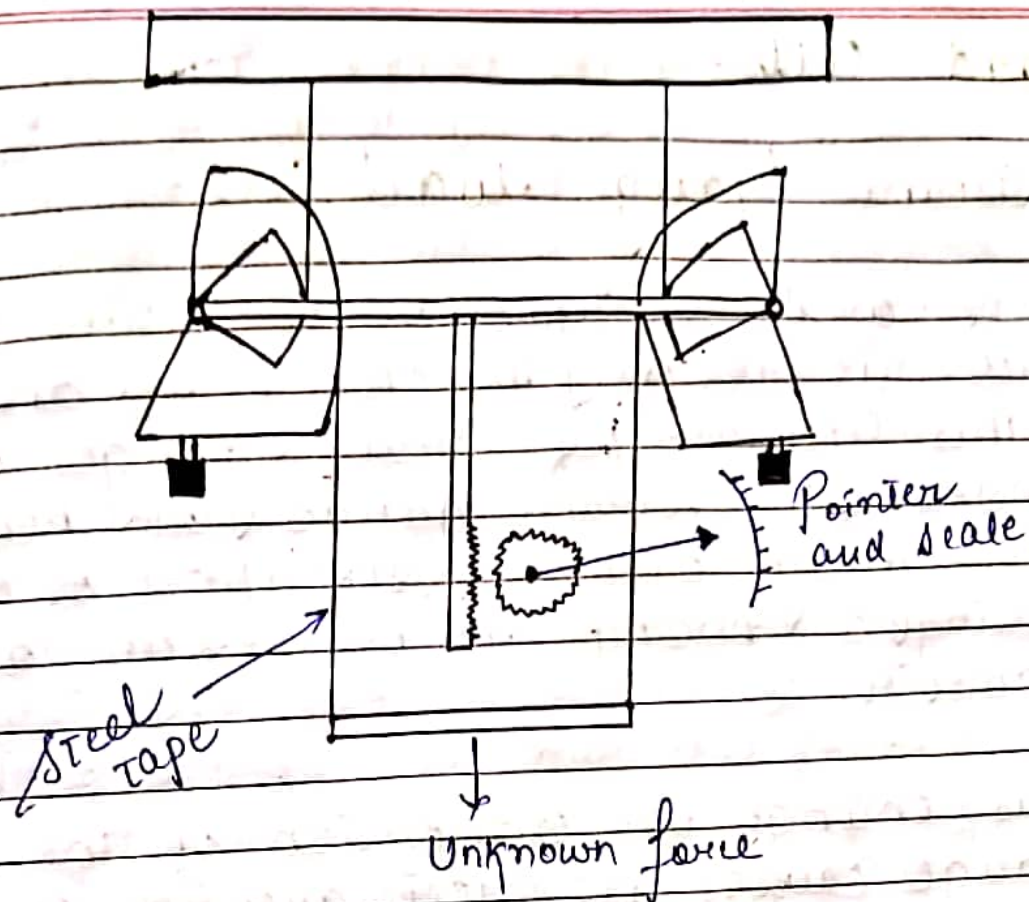
- Unequal arm balance - An Unequal arm balance work on the Principle of moment Comparison.

The beam of the unequal arm balance is in equilibrium position when,

Clockwise Rotating moment = Anti-Clockwise Rotating moment.

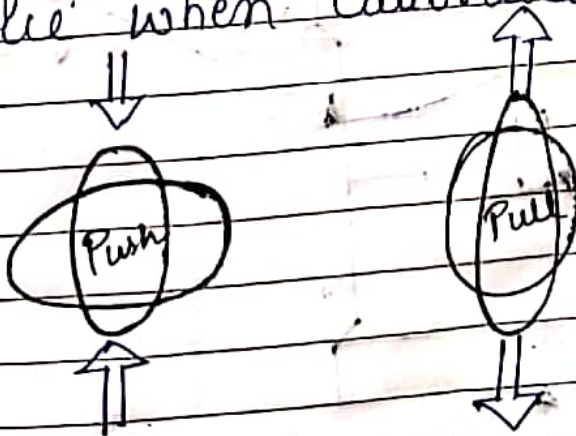


- Pendulum Scale (Multi-level Type).
It is a moment comparison device. The Unknown force is converted to Torque which is then balanced by the Torque of a fixed standard Mass arranged as a Pendulum.



Elastic force meter (Proving Ring) -

When a steel ring is subjected to a force across its diameter, it deforms. This deflection is proportional to the applied force when calibrated.



• Load cells -

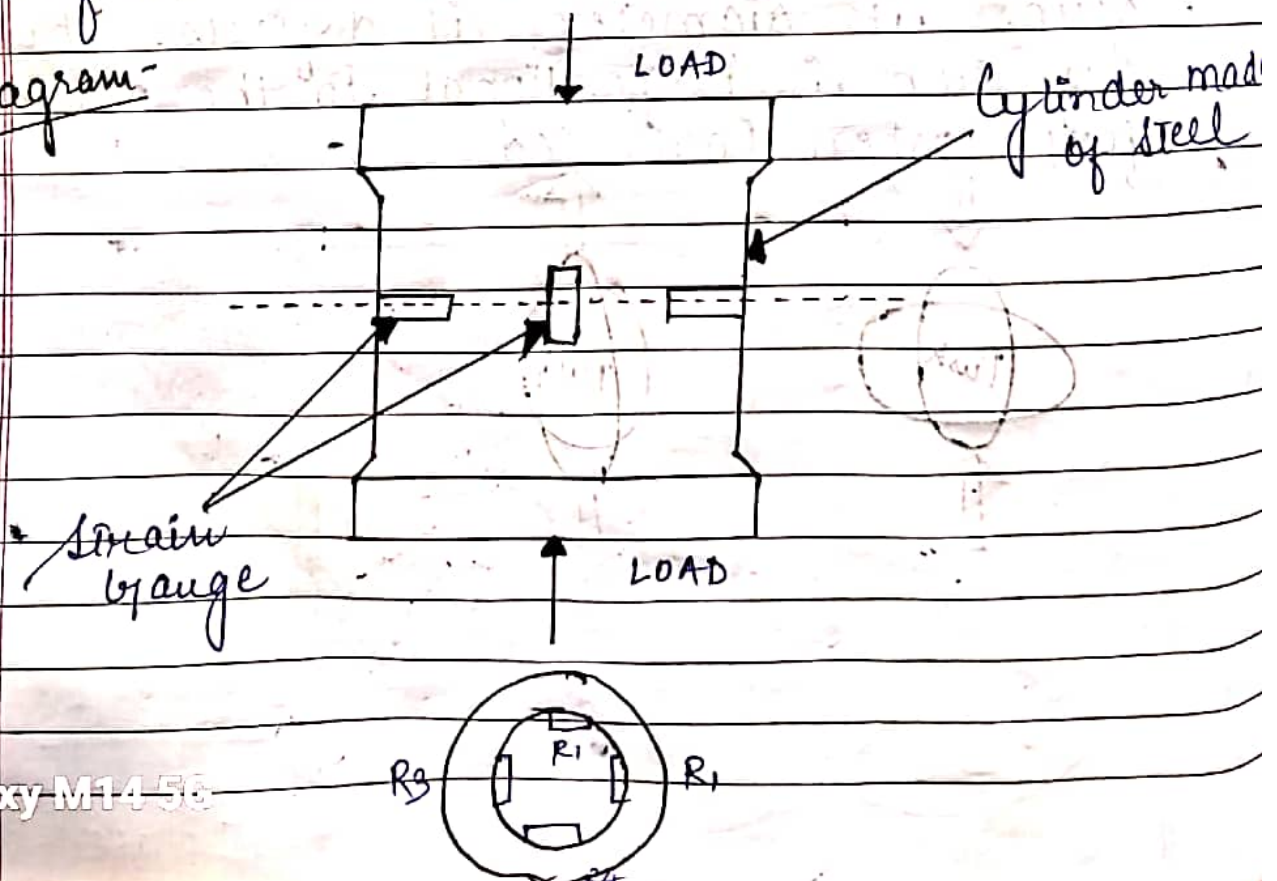
• Strain Gauge Load Cell -

• When a steel cylinder is subjected to a force, it tends to change in dimension. On this cylinder if strain gauges are bonded, the strain gauge is stretched or compressed causing a change in its length and diameter.

• This change in dimension of the strain gauge causes its resistance to change.

This change in resistance of the strain gauge becomes a measure of the applied force.

Diagram -

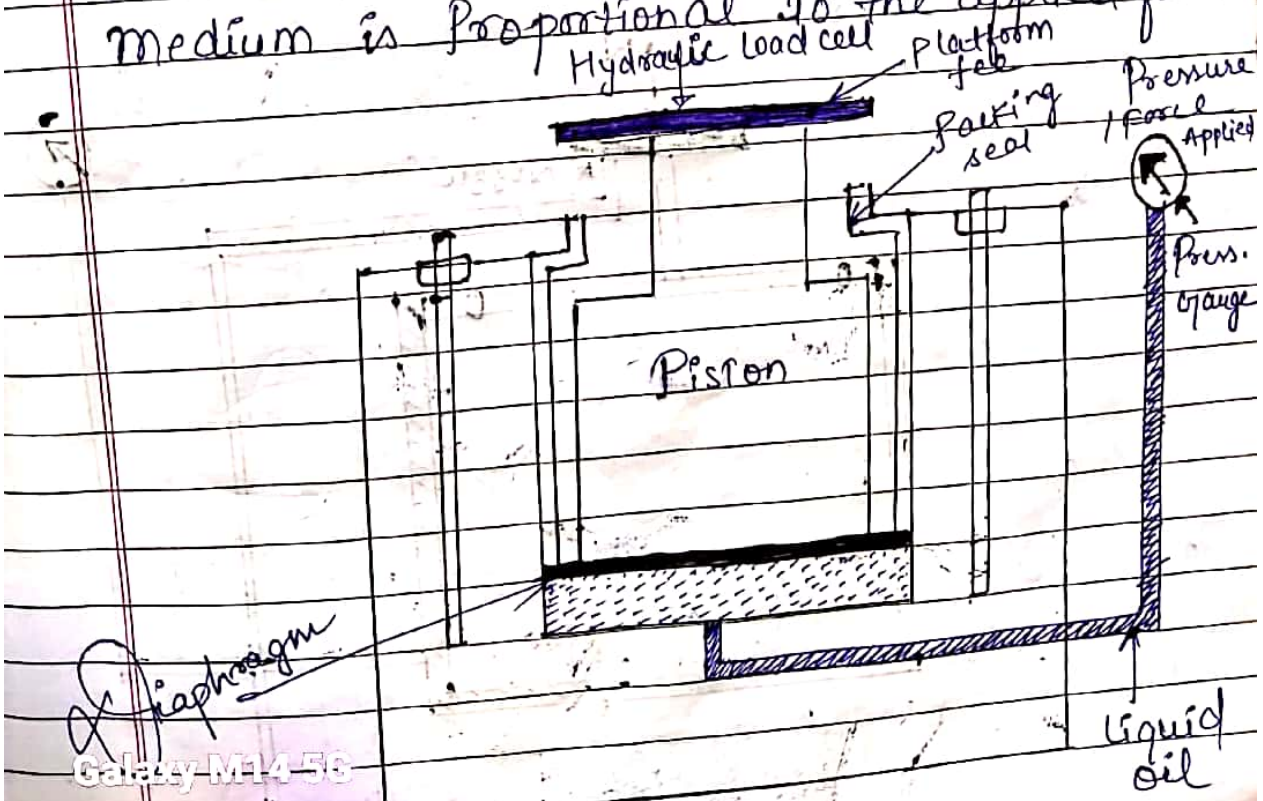


Hydraulic load cell-

When a force is applied on liquid medium contained in a confined space, the pressure of the liquid increases. This increase in pressure of the liquid is proportional to the applied force.

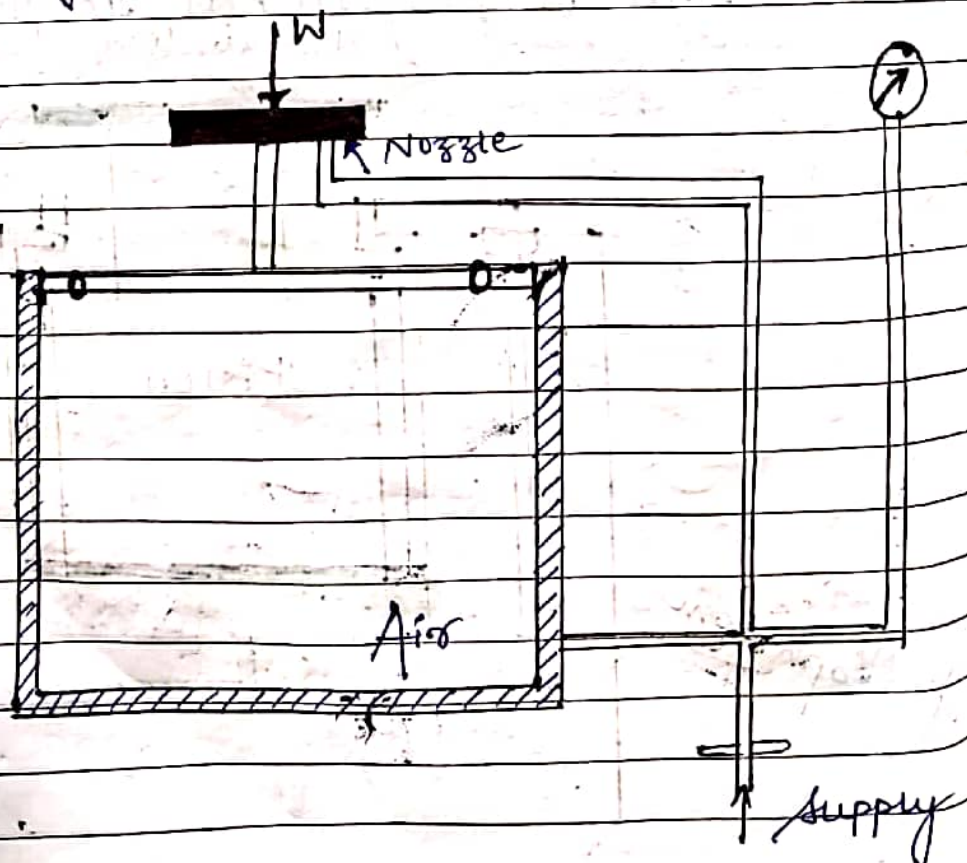
The force to be measure is applied to the Piston.

- The applied force moves the Piston downwards and deflects the diaphragm and this deflection of the diaphragm increases the Pressure in the liquid medium.
- This increase in Pressure of the liquid medium is proportional to the applied force.



c) Pneumatic load cell -

- If a force is applied to one side of a diaphragm and an air pressure is applied to the other side, some particular value of pressure will be necessary to exactly balance the force. This pressure is proportional to the applied force.
- The force to be measured is applied to the top side of the diaphragm. Due to this force, the diaphragm deflects and causes the flapper to shut-off the nozzle opening.
- Air supply is provided at the bottom of the diaphragm. As the flapper closes the nozzle opening, a back pressure results underneath the diaphragm.



Elastic Force Transducer

Force sensor is a type of transducer, specially a force transducer. It converts an input mechanical force such as load, weight, tension, compression or pressure into another physical variable, in this case, into an electrical output signal that can be measured, converted and standardized. As the force applied to the force sensor increases, the electrical signal changes proportionally.

Transducers became an essential element in many industries from Automotive, High Precision Manufacturing, Aerospace, Defense, Industrial Automation, Medical & Pharmaceuticals and Robotics where reliable and high precision measurement is paramount.

How does it work - It works on the principle of strain gauge (sometimes referred to as strain gauge).

Metal foil strain gauge is a sensor whose electrical resistance varies with applied force. In other words, it converts (or transducers) force, pressure, tension, compression, torque, etc. into a change in electrical resistance which can then be measured.

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LOAD CELL

A load cell is a Transducer that converts the mechanical force into standard electrical units, similar to our regular weighing scales. The main purpose is to weigh or check the amount of load transferred.

The load cell sensors are always bonded along with elastic material, known as strain gauges.

Used $\rightarrow$ load cells can be used at the initial stage i.e. during the research and development or even at a later stage while monitoring a structure.

Geotechnical instrument are used both pre and post-construction to ensure the safety of structure, dams, tunnels, bridges etc. Proper geotechnical monitoring ensures the long-term safety of these structure.

Load cells find their application in the geotechnical field and they are commonly used to monitor -

*) Deep foundations :- Excavation bracing like struts, Soldier Pile, Tiebacks of anchors, retaining walls.

- 2) Tunnels and shafts :- steel liner Plate, Cast-in-Place concrete, Segmented Precast concrete, shotcrete.
- 3) Dams - Concrete Dams, Underground Power House.
- 4) Piles - Pile load Test.

Working Principle of a load cells:-

The load Cell Principle involves the use of many specific geotechnical instruments. It can't work without being Paired up with sensors, one of them being strain gauges.

Strain gauges are thin elastic materials made up of stainless steel and are fixed inside the load cells using Proprietary adhesives.

Types of load cells -

- 1) Model ELC - 150S - H High Capacity Compression load cell
- 2) Model ELC - 210S Resistive strain Gauge Type Compression load cell
- 3) Model ELC - 30S Resistive strain Gauge Type Center Hole.
- 4) Model ELC - 30S - H Resistive strain Gauge Type

5) MODEL ELC - 31 V vibrating Wire Hydraulic Center Hole - Anchor Bolt Load Cell.

6) Model ELC - 32 V vibrating Wire Center Hole - Anchor Bolt Load Cell

7) Piezoelectric Load Cell

Applications of Load Cell -

- ① To determine the load in rock bolts, Tiebacks, foundation anchors, Cables, or struts.
- ② Proof Testing and long-Term Performance monitoring of different kinds of anchor systems.
- ③ Compressive load measurement between structural members i.e Tunnel Supports or at the junction between a beam and the top of a pile strut.
- ④ Load Testing in Piles
- ⑤ Measurement of Compressive load.

Shear Web/Pan Cake Load Cell

The Shear Web/Pan Cake Load cell (also known as the Universal Load cell) category is specifically engineered for Endurance or Inline application.

These load cells, with their low-profile and donut-style shape, are favored for force measurement application due to their ability to be installed within limited real estate easily.

Shear Web load cells feature a Central threaded hole that facilitates measuring loads in both Tension and Compression. Their Unique Capability to Compensate for off-axis loading, also termed off-center loading, without compromising accuracy makes them the Preferred choice for applications that demand high endurance fatigue life, or high-capacity in-line measurements. These attributes render shear web load cells ideal for diverse applications such as Tank weighing systems, Tablet Compression force analysis, and Injection molding clamping force monitoring.

Pancake load cells are equipped with bonded foil strain gauges that enables them to withstand challenging environments including high-Temp. Setting.

Offering an accuracy range of 0.025% to 0.10% for full-scale output, Pancake load cells are available with high-level analog voltage, analog current, or digital output options to suit the buyer's requirements. Most high-level Pancake load cells are integrated with internal signal conditioners featuring Proprietary electronics. Irrespective of the output type, these load cells are designed with some form of the standard electrical receptacle for seamless integration.

Pancake load cells are designed in various configurations such as high-accuracy, Tension-Compression; fatigue rated, Precision and Coiled Tubing injector Pancake load cells.

Piezoelectric Force Transducer

The Piezoelectric force sensor, also known as a load washer or measurement washer, is the classic piezoelectric measurement element for the measurement of force along a single axis. It describes either a compression or tensile force F_z or shear force $F_{x,y}$.

How are Piezoelectric force sensor used?

Piezoelectric force sensors are normally integrated at the measuring point directly in the force flux of the existing structure and installed and calibrated with the required pretension.

Pretensioned piezoelectric force sensors or force transducers are immediately ready to perform measurements and are delivered calibrated. They can be used to measure compression and tensile forces.

Piezoelectric force sensors are used in various applications, for example, in mounting technology, in spot welding or in presses, during the measurement of impact resistance, cutting forces and forming forces or in force plates, weigh in motion systems and in crash-test

Strain Gauge



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A strain gauge is one of the imperative devices used in the field of geotechnical engineering to measure strain on diverse structure. By applying an external force, there would be a change in resistance of a strain gauge.

The basic construction of a gauge has an insulating flexible backing. To support a metallic foil structure. This metallic coil is glued to a thin backing called a carrier, and the entire setup is fixed to an object using a suitable adhesive.

As the object is deformed due to force, Pressure, Weight, Tension, etc., the electrical resistance of foil changes.

A wheatstone bridge measures the change in resistivity, which is related to strain through a quantity known as gauge factor.

Strain Gauge Working Principle:-

Strain Gauge Working Principle is based on the concept of electrical resistance variation due to mechanical strain. The strain gauge consists of a fine wire or foil, typically made of a highly conductive

Material like copper or constantan.

Applications of Strain Gauge:-

- The applications of strain gauges in the field of Civil Engineering are invaluable!
- Strain gauge are extensively used to measure and monitor the deformations and stresses experienced by various structural components and materials.
- One significant applications of strain gauge is in the field of structural health monitoring. By attaching strain gauges to critical sections of structures such as bridges, dams and buildings etc.
- Strain gauges are also used in experimental studies and material testing.
- Strain Gauges also have applications in geotechnical engineering. They are employed to monitor the deformations in soil, rock and underground structures such as tunnels and retaining walls.

Advantages -

- ① High Sensitivity
- ② Versatility
- ③ Cost - Effective
- ④ Real - Time Monitoring

Disadvantages -

- ① Installation Complexity -
- ② Susceptible To Environmental factors.
- ③ Limited Measurement Range.
- ④ Vulnerability To Damage.

Inductive Transducer

Inductive Transducer is the self-generating type otherwise the Passive Type Transducer.

The first type like self generating uses the principle of fundamental electrical generator. The electric generator principle is when a motion among a conductor as well as magnetic field induces a voltage within the conductor.

Working Principle :-

The Working Principle of an Inductive Transducer is the magnetic material induction. The magnetic material's induction can depend on different variables like the twist of the coil over the material, the magnetic material's size & the flux permeability.

The magnetic materials are used in the transducer in the path of flux. There is some air gap between them. The change in the circuit inductance can be occurred due to the air gap change.

Magnetostrictive Transducer

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Inductive Transducer operate on the Principle of Variation of self inductance of a coil or on the Principle of variation of mutual inductance.

Inductance of a coil, L is defined as -

$$L = \frac{N d\phi}{di}$$

Assuming linear relationship between flux, ϕ current, i producing the flux, we can write -

$$L = \frac{N\phi}{i} = \frac{NBA}{i} = \frac{N\mu H A}{i}$$

[as $\phi = B \cdot A$
and $B = \mu H$]

$$L = \frac{N \cdot \frac{\mu N i}{l} \cdot A}{i}$$

$$\left[H = \frac{NI}{l} \right]$$

$$L = \frac{\mu N^2 A}{l}$$

where, μ = Permeability of the core material

N = Number of turns of the coil.

A = Area of cross-section of magnetic flux Path.

l = length of flux Path.

Thus, the inductance of a coil can be varied by -

- ① changing the Permeability of the core material
- ② Changing the number of turns through which current will flow in the coil.
- ③ Changing the ratio of $\frac{A}{l}$ of the coil.

Two coils having inductance L_1 and L_2 will have mutual inductance M as -

$$M = K \sqrt{L_1 L_2}$$

where $K \rightarrow$ Coefficient of Coupling

Magnetostrictive Transducer

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~~The electrochemical transducer is a device used to convert etc~~

A device that is used to change the energy from Mechanical to magnetic Energy is known as a magnetostrictive transducer.

Working - The Working of a magnetostrictive transducer can be described by using the following schematic diagram.

This diagram explains the strain amount produced from null to complete magnetization. This is divided into discrete mechanical & magnetic attributes that are set in their effect on the magnetic induction & magnetostrictive core strain.

Types of Magnetostrictive Transducer

There are two types of magnetostrictive transducers, spontaneous magnetostriction and field-induced magnetostriction.

Spontaneous Magnetostriction - Spontaneous magnetostriction occurs from the magnetic ordering of atomic moments under the Curie temperature.

② Field-Induced Magnetostriction -
field-Induced magnetostriction main
curves mainly from the magnetic
domain arrangement on an applied
field applications.

How to choose Magnetostrictive
Transducer?

- This transducer must use a type of
magnetic material so that it can
interact and can map distances very
exactly.
- Transducers must allow contact-free
wear-free measurements.
- Its range must be from 50 to 2500 mm
- Its maximum resolution should be approx
2 μm .
- Maximum linearity must be $\pm 0.01\%$.
- Displacement speed should be less than
10 m/s.
- Analog output is 0 to 10V, 4 to 20 mA
- 24 V DC $\pm 20\%$ voltage supply.

Advantages -

These transducers are reliable, maintenance-free, significantly reduce the potential for operating errors & machine downtime.

Magnetostrictive Transducers don't have Contact Parts, so they have a longer life.

These are more accurate as compared to fixed contact transducers.

They have Good Sensitivity, long-range inspection, durability, easy implementation etc.

Disadvantages -

Magnetostrictive Transducers are Expensive.

The magnetostrictive Transducers has Physical size limitations, so it is restricted to operate at below 30 KHz frequencies.

Applications -

It is used for Position Measurement.

It Plays a key role in converting Mechanical Energy into Magnetic Energy.

It is used to Make different devices like high-torque linear Motors, Noise control System.

It is used to make Surgical Tools, Chemical